

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_v^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & 0 \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxzz} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)}$$

$$M_{yzyz} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & 0 \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = -\mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2\mathbb{M}_v^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzzy} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{zzxx} = -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{yzyz} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzzy} = \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzy} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

$$M_{yzzy} = -2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

$$M_{xyxy} = 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & 0 & 0 \\ 0 & -M_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{yyzz} \\ M_{zzxx} & 0 & 0 \\ 0 & -M_{yyzz} - M_{zzxx} & 0 \\ 0 & M_{yzyy} & 0 \\ 0 & 0 & M_{zzxz} \\ -M_{yzyy} - M_{zzxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{xxxx} &= \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)} \\ M_{yyzz} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\ M_{zzxx} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\ M_{yzyy} &= \mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\ M_{zzxz} &= -\mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \end{aligned}$$